Maths

Q1 monomial

Q2 $(x+1)(x^2-x-x^4+1)$

$x(x^2-x-x^4+1)+1(x^2-x-x^4+1)$

$x^3-x^2-x^5+x+x^2-x-x^4+1$

$-x^5-x^4+x^3-x+1$

$\therefore$ degree of polynomial $= 5$

Q3 If 1 is zero, then, ~~f(x)=~~ $f(1)=0$

$\therefore f(x)=a^2x^2-3ax+3x-1$

$f(1)=a^2(1)^2-3a(1)+3(1)-1$

$= a^2-3a+3-1$

$0 = a^2-3a+2$

~~$-2 = a^2-3a$~~

$0 = a^2-2a-a+2$

$= a(a-2)-1(a-2)$

$0 = (a-1)(a-2)$

$\therefore a=1$ and 2

Q4 Since the zeroes are the points on the graph where it intersects x-axis, ~~this ha~~ $p(x)$ has 1 zero.

Q5 Let α be the first zero of $2x^2-3x+k$

$\therefore$ other zero $= \frac{1}{\alpha}$

Now, product of ~~zeo~~ zeroes $= \frac{c}{a}$

In $2x^2-3x+k$

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$a = 2,\ b = -3,\ c = k$

$\therefore\ \cancel{\alpha} \times \frac{1}{\cancel{\alpha}} = \frac{k}{2}$

$1 = \frac{k}{2}$

$2 = k$

$\therefore k = 2$

Q6 Sum of zeroes $= \alpha + \beta = 5$
Product of zeroes $= \alpha\beta = -14$

So, new quadratic equation $= x^2 - (\alpha+\beta)x + \alpha\beta$
$= x^2 - (5)x + (-14)$
$= x^2 - 5x - 14$

Q7 $x^2 + 5x + 6 \Rightarrow x^2 + 2x + 3x + 6$
$\Rightarrow x(x+2) + 3(x+2)$
$\Rightarrow (x+3)(x+2) = 0$

$\therefore x = -3$ or $x = -2$
Let $\alpha = -3$, $\beta = -2$.
where α and β are zeroes of x^2+5x+6

Now, $\alpha + \beta = \frac{-b}{a}$

In $x^2 + 5x + 6$
$a = 1,\ b = 5,\ c = 6$

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$\therefore \alpha+\beta = \cancel{-8}\ \frac{-b}{a}$

LHS, $\alpha+\beta = -3-2$
$= -5$

RHS, $\frac{-b}{a} = \frac{-5}{1}$

$= -5$

$-5 = -5$

$\therefore$ Verified

Now, $\alpha\beta = \frac{c}{a}$

LHS, $\alpha\beta = (-2)\times(-3)$
$= +6$

RHS, $\frac{c}{a} = \frac{6}{1}$

$= 6$

$6 = 6$

$\therefore$ Verified

Q8 Given zeroes, 3, 5, −2

therefore the cubic equation $= (x-3)(x-5)(x+2)$
$= (x^2-8x+15)(x+2)$
$= x^3-6x^2-x+30$

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Q9 ~~If~~ We know, $x+\frac{1}{x}=3$

Now, $\left(x+\frac{1}{x}\right)^3 = x^3+\frac{1}{x^3}+3\left(\cancel{x}\times\frac{1}{\cancel{x}}\right)\left(x+\frac{1}{x}\right)$

$(3)^3 = x^3+\frac{1}{x^2}+3(\cancel{x}\,3)$

$27 = x^3+\frac{1}{x^3}+9$

$18 = x^3+\frac{1}{x^3}$

Q10 ~~$15x^2-5x+6 = 15x^2-3x-2x+6$~~
~~$= 3x(5x-1)+2$~~

In ~~$15x^2-3x-2x+6$~~ $15x^2-5x+6$

$a=15,\ b=-5,\ c=6$

$\therefore \alpha+\beta = \frac{-b}{a} = \frac{-(-5)}{15} = \frac{1}{3}$

$\therefore \alpha\beta = \frac{c}{a} = \frac{6}{15} = \frac{2}{5}$

Now, $\left(1+\frac{1}{\alpha}\right)\left(1+\frac{1}{\beta}\right) = \left(\frac{\alpha+1}{\alpha}\right)\left(\frac{\beta+1}{\beta}\right)$

$= \frac{(\alpha+1)(\beta+1)}{\alpha\beta}$

$= \frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$

$= \frac{\frac{2}{5}+\frac{1}{3}+1}{\frac{2}{5}}$

$$= \frac{\frac{6+5+15}{15}}{\frac{2}{5}}$$

$$= \frac{\frac{26}{15}}{\frac{2}{5}} = \frac{\overset{13}{\cancel{26}} \times \overset{1}{\cancel{5}}}{\cancel{2} \times \underset{3}{\cancel{15}}} = \frac{13}{3}$$